

Model Backbone Sensitivity Analysis

This document examines whether the proposed alignment algorithms remain effective when evaluated with different LLM backbones. First, we report the performance of Llama-4-Maverick-17B-128E across different alignment algorithms on the World Values Survey dataset (Table D.12). Second, to address the potential self-reinforcement concern, we evaluate GPT-4o-crafted system prompts using different inference models, including GPT-4o, Llama-4-Maverick, and Llama-3.1-70B, on the MobLab dataset (Table D.13).

Table D.12. Distributional alignment performance of Llama-4-Maverick-17B-128E on the WVS dataset across EM and GB algorithms.

Algorithm	W-dist	Mean	Std
EM	1.55	43.83	18.16
GB	1.47	44.62	18.98

Table D.13. Cross-model backbone sensitivity analysis on the MobLab dataset. System prompts are crafted by GPT-4o and evaluated using different inference models.

Inference Model	Method	Dictator	Proposer	Responder	Investor	Banker	Public Goods	Bomb
GPT-4o	EM	1.69	1.39	3.05	1.75	9.36	0.47	6.32
	GB	1.17	1.88	2.51	1.84	4.24	0.88	4.59
Llama-4-Maverick	EM	1.94	2.95	18.68	3.63	14.48	0.59	13.00
	GB	3.73	3.87	12.43	5.01	6.16	1.59	10.74
Llama-3.1-70B	EM	1.60	7.57	4.49	5.55	9.37	0.73	5.37
	GB	4.10	7.11	4.43	12.43	6.80	2.55	7.07